

The fourfold Chang cylinder is unstable for symmetric Gaussian perimeter

Candidate full solution packet for arXiv:2204.12003

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Status: candidate full solution, likely valid pending expert review.

Abstract

Heilman left open the Gaussian-perimeter instability of the noncircular convex λ -shrinker curves with rotational orders $m = 2$ and $m = 4$. We settle the $m = 4$ case. In fact, every Chang curve of even rotational order $m \geq 4$, and every Euclidean cylinder over it, is unstable among centrally symmetric competitors of fixed Gaussian volume. The proof is a nodal-count correction: the normal component of infinitesimal rotation has two nodal domains per curvature period, hence $2m$ on the closed curve. For $m = 4$ there are eight, so Heilman's existing "more than four nodal domains" criterion applies. The same argument reaches exactly four domains for $m = 2$ and therefore does not settle that remaining case.

1 The source question

The source is S. Heilman, *Convex Cylinders and the Symmetric Gaussian Isoperimetric Problem*, arXiv:2204.12003 [1]. Its page 4 recalls the embedded convex curves Γ_m constructed by Chang [2], which after normalization satisfy

$$H = \langle x, N \rangle + \lambda, \quad \lambda < 0, \quad (1)$$

and have minimal rotational symmetry $2\pi/m$. The source invokes a nodal criterion for even $m \geq 6$, then says that instability is unclear for $m = 2$ and $m = 4$.

with respect to a rotation by an angle smaller than $2\pi/m$) [Cha17, Theorem 1.2, Theorem 1.3, Proposition 3.2]. Consequently, $\Gamma_m \times \mathbb{R}^{n-2} \subseteq \mathbb{R}^{n+1}$ also satisfies $H(x) = \langle x, N(x) \rangle + \lambda$. So, Huisken's classification cannot possibly hold, at least when $\lambda < 0$.

In the case of even $m \geq 6$, the curve Γ_m can be shown to be unstable by [Hei21, Corollary 11.9], since there are more than four nodal domains corresponding to an infinitesimal rotation of Γ_m . However, it is unclear how to prove instability for the cases $m = 2$ and $m = 4$. As shown at the end of [Cha17], there appears to be an infinite family of curves with a single symmetry by a rotation of an angle π , and it is unclear if any of our arguments can show these curves are unstable for the Gaussian perimeter.

1.3. Future Directions and Remaining Cases of Barthe's Problem. The ultimate goal of Barthe's Problem [Bar11] is to find the minimum Gaussian perimeter of a set in $\mathbb{R}^{n+1}$

Figure 1: The unresolved $m = 2$ and $m = 4$ statement on page 4 of arXiv:2204.12003.

Write

$$\gamma_j(x) = (2\pi)^{-j/2} e^{-|x|^2/2}.$$

Instability below means that there is a centrally symmetric, Gaussian-volume-preserving normal variation with negative second variation of Gaussian perimeter.

2 Proof intuition

There are two ingredients, both already visible in the cited papers. First, the normal speed generated by rotating a critical surface is a zero mode of the Gaussian Jacobi operator. Heilman's nodal argument says that a centrally symmetric minimizer cannot have such a rotation mode with more than four nodal domains. Second, Chang's phase variable $\tau = \langle x, T \rangle$ is, up to sign, exactly that rotation mode, and $k' = k\tau$. A nonconstant curvature oscillation has one increasing and one decreasing half, hence two nodal domains of τ . Chang's m -fold closing condition repeats the full oscillation m times. The total is therefore $2m$, giving eight domains when $m = 4$. The source's existing nodal criterion then finishes the proof without a new spectral estimate.

3 A self-contained nodal criterion

We first recall the mechanism behind Corollary 11.9 of [3]. Let $\Sigma = \partial\Omega \subset \mathbb{R}^q$ be a smooth critical point of Gaussian perimeter under a Gaussian-volume constraint, and let

$$L = \Delta_\Sigma - \langle x, \nabla_\Sigma \rangle + |A|^2 + 1 \quad (2)$$

be its Gaussian Jacobi operator. Put

$$\mathcal{B}(u, v) = \int_\Sigma uLv \gamma_{q-1}.$$

With this convention, a symmetric volume-preserving normal speed u has second variation $-\mathcal{B}(u, u)$. Thus stability requires $\mathcal{B}(u, u) \leq 0$ whenever $u(-x) = u(x)$ and $\int_\Sigma u \gamma_{q-1} = 0$.

Lemma 1 (rotation nodal criterion). *Assume $\Omega = -\Omega$. Let Q be a nonzero antisymmetric matrix and set*

$$f(x) = \langle Qx, N(x) \rangle.$$

If f has more than four nodal domains, then Ω is unstable for the symmetric Gaussian isoperimetric problem.

Proof. Rotating (1) and differentiating at the identity gives $Lf = 0$. Moreover $f(-x) = f(x)$, and the weighted divergence theorem gives

$$\int_\Sigma f \gamma_{q-1} = \int_\Omega \operatorname{div}(\gamma_q Qx) = \int_\Omega (\operatorname{tr} Q - \langle x, Qx \rangle) \gamma_q = 0.$$

Suppose for contradiction that Ω is stable. Antipodal symmetry pairs the nodal domains of f . More than four domains allow us to choose two different antipodal pairs $D_1 \cup (-D_1)$ and $D_2 \cup (-D_2)$ while leaving an unused nodal domain. Define g to equal f on the first pair, αf on the second pair, and zero elsewhere. Because f has one strict sign on each nodal domain, one can choose $\alpha \in \mathbb{R}$ so that $\int g \gamma_{q-1} = 0$. Nodal-domain integration by parts and $Lf = 0$ give $\mathcal{B}(g, g) = 0$.

For every smooth symmetric mean-zero h , stability applied to $g + th$ for all $t \in \mathbb{R}$ now forces $\mathcal{B}(g, h) = 0$. Equivalently, Lg is a constant distribution in the symmetric sector. On the unused open nodal domain, g vanishes identically, so that constant is zero. Elliptic regularity and unique continuation then imply $g = 0$ everywhere, contradicting its definition. The nodal truncations used here belong to the Gaussian $W^{1,2}$ space and are obtained as limits of smooth speeds; this is the standard justification given in [3, Corollary 11.9]. $\square$

4 The missing nodal count

Let Γ_m be one of Chang's noncircular embedded convex solutions, parametrized by arclength s . Choose the orientation so that its curvature $k = H$ is positive, let T be the unit tangent, and put

$$\tau = \langle x, T \rangle.$$

Differentiating (1) and the Frenet equations gives the phase-plane system

$$k' = k\tau, \quad \tau' = 1 + \lambda k - k^2. \quad (3)$$

These are Chang's equations after the normalization used in arXiv:2204.12003.

Lemma 2. *The normal component of infinitesimal rotation on Γ_m has exactly $2m$ nodal domains.*

Proof. Let Q be rotation through $\pi/2$. Depending only on the orientation,

$$\langle Qx, N \rangle = \pm \langle x, T \rangle = \pm \tau.$$

It therefore suffices to count the sign intervals of τ .

Chang writes $B = 2 \log k$ and reduces (3) to the energy equation

$$\frac{1}{2}(B')^2 + 2V(B) = 2\eta, \quad V(B) = e^B - 2\lambda e^{B/2} - B. \quad (4)$$

For $\lambda < 0$, V is strictly convex. Every nonconstant periodic orbit therefore has exactly two turning values $B_- < B_+$, and B' has one strict sign between successive turning points. Since $B' = 2k'/k = 2\tau$, the function τ has exactly two nodal domains during one full phase period. The two zeros are simple: if $\tau = \tau' = 0$, then (3) is at its equilibrium, and uniqueness would make the curve circular.

In Chang's closing construction, one full phase period changes the tangent angle by $2\pi/m$; repeating it m times closes Γ_m and gives its minimal m -fold rotational symmetry. Consequently the two sign intervals per phase period produce exactly $2m$ nodal domains on the closed curve. $\square$

For comparison, if θ denotes normal angle, then $d\theta/ds = k$ and (3) becomes

$$k_{\theta\theta} + k = \lambda + \frac{1}{k}. \quad (5)$$

On functions of the curve, (2) becomes

$$Lf = f + k^2(f_{\theta\theta} + f).$$

Differentiating (5) verifies directly that $L(k_\theta) = 0$, while $k_\theta = k'/k = \tau$. This independently checks both the Jacobi field and its two zeros per curvature period.

5 Result

Theorem 3. *Let $\Gamma_m \subset \mathbb{R}^2$ be a noncircular embedded convex Chang curve with rotational order m , normalized by (1). If m is even and $m \geq 4$, the centrally symmetric region bounded by Γ_m is unstable for Gaussian perimeter under centrally symmetric, Gaussian-volume-preserving variations. More generally, for every $d \geq 0$, the cylinder*

$$\Gamma_m \times \mathbb{R}^d \subset \mathbb{R}^{d+2}$$

is unstable in the same sense. In particular, this settles the $m = 4$ case identified as open in the source paper.

Proof. Even m makes the bounded region centrally symmetric. By Lemma 2, the infinitesimal-rotation Jacobi field has $2m$ nodal domains. For $m \geq 4$, this number is at least eight and hence is strictly larger than four. Lemma 1 proves instability.

On $\Gamma_m \times \mathbb{R}^d$, rotate only the first two coordinates. The same Jacobi field is independent of the $\mathbb{R}^d$ coordinate, and each nodal domain is a nodal arc times the connected space $\mathbb{R}^d$. Thus its number of nodal domains is still $2m$. Gaussian integrals factor, so Lemma 1 applies without change. $\square$

Remark 4. For $m = 2$, Lemma 2 gives exactly four nodal domains. The strict “more than four” hypothesis in Lemma 1 is then unavailable. The present proof therefore resolves the source’s $m = 4$ case but does not claim a solution of the $m = 2$ case.

6 Novelty check

The run’s cheap indexes contained no entry for arXiv:2204.12003 or the low-symmetry λ -shrinker question. Bounded arXiv searches on 9 August 2026 used the source title and identifier together with “ $m = 4$ ”, “ Γ_4 ”, “four nodal domains”, “infinitesimal rotation”, and “Gaussian perimeter instability”. They found the three sources below but no later paper recording the $2m$ count or settling the $m = 4$ case. This is evidence of possible novelty, not an exhaustive bibliographic guarantee.

References

- [1] S. Heilman, *Convex Cylinders and the Symmetric Gaussian Isoperimetric Problem*, arXiv:2204.12003 (2022).
- [2] J.-E. Chang, *One dimensional solutions of the λ -self shrinkers*, arXiv:1410.1782; *Geom. Dedicata* (2017), 1–16.
- [3] S. Heilman, *Symmetric Convex Sets with Minimal Gaussian Surface Area*, arXiv:1705.06643; *Amer. J. Math.* 143 (2021), 53–94.